

SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed-to-be-University under Section 3 of UGC Act, 1956)

Cloud Security Fundamentals

Protecting Data and Applications

Technical Presentation

CSA1512-CLOUD COMPUTING AND BIG DATA ANALYTICS

DATA

CLOUD SECURITY

APPS

GUIDED BY:

Dr.R.LATHA

PRESENTED BY:

R.YUVASHREE(192521485)

Objectives

What this presentation covers

- Protect cloud data and applications
- Prevent unauthorized access and cyberattacks
- Understand major cloud security models
- Learn shared responsibility
- Understand access control and encryption
- Learn monitoring and backup practices

Core Security Models

Use multiple layers instead of relying on one control

- CIA Triad – Confidentiality, Integrity, Availability
- Zero Trust – Never trust, always verify
- Defense in Depth – Use multiple security layers
- Protect data from unauthorized access
- Verify users before giving access
- Reduce security risks through layered protection

Essential Cloud Security Controls

Controls should protect identity, network, application, and data

- IAM – Controls user access
- MFA – Provides extra login security
- Encryption – Protects sensitive data
- Firewall – Controls network traffic
- WAF – Protects web applications
- Monitoring – Detects suspicious activities
- Backup & Recovery – Restores data after failures or attacks

Secure Cloud Application Architecture

- User Authentication – Verify users using IAM and MFA.
- Network Security – Use firewalls and WAF to block threats.
- Application Security – Protect applications from vulnerabilities.
- Data Protection – Use encryption for sensitive data.
- Access Control – Give users only required permissions.
- Monitoring – Track activities and detect suspicious behavior.
- Backup & Recovery – Restore data during failures or attacks.
- Multiple Security Layers – Protect the application at every stage.

Secure Access Process

How a typical request is validated before reaching protected data

1

Authenticate

User signs in with password + MFA.

2

Authorize

IAM checks role, policy, device and context.

3

Protect

TLS encrypts traffic; WAF filters malicious requests.

4

Process

Application validates input and accesses only required resources.

5

Record

Audit logs capture identity, action, timestamp and result.

6

Recover

Backups and tested recovery procedures limit impact after failure.

Real-World Use Case: Online Banking

Goal: protect accounts, transactions, customer data, and service availability

Threats

- Credential theft / account takeover
- SQL injection & XSS
- DDoS and bot traffic
- Data leakage
- Insider misuse
- Ransomware / service outage

Controls

- MFA + risk-based IAM
- WAF + input validation
- Network segmentation
- Encryption + managed keys
- Central audit logging
- Immutable backups + DR

Outcome

Unauthorized access is harder to achieve, sensitive data remains protected, suspicious activity is detected, and the service can recover quickly after an incident.

Real-World Security Practices

Security is a continuous operational process

- Use least privilege and MFA for human and machine identities.
- Store secrets in a dedicated secrets manager—not source code or plain-text configuration.
- Encrypt sensitive data in transit and at rest; rotate and tightly control keys.
- Patch dependencies and operating systems; continuously scan infrastructure and containers.
- Centralize logs and monitor high-risk events such as privilege changes and unusual access.
- Test backups, incident-response procedures, and disaster recovery—not just the configuration.
- Regularly review cloud permissions and remove unused accounts, keys, ports, and resources.

Conclusion & Key Takeaways

A secure cloud environment is designed, monitored, and continuously improved

1. Security is layered

Use Zero Trust, defense in depth, IAM, network controls, encryption, application security, monitoring, and recovery together.

2. Responsibility is shared

The provider secures the cloud infrastructure; the customer remains responsible for many configuration, identity, data, and application controls.

3. Security is continuous

Prevent → detect → respond → recover. Regular testing and monitoring are essential.